💳 Credit Risk Assessment & Loan Default Analysis

📌 Project Overview

This project presents an end-to-end **Credit Risk Assessment & Loan Default Analysis** using **SQL** and **Microsoft Power BI**. The objective is to analyse customer credit profiles, evaluate loan approval trends, identify high-risk borrowers, and provide actionable insights that support responsible lending decisions.

Using SQL for data extraction and Power BI for interactive visualisation, this project demonstrates how financial institutions can leverage data analytics to minimise credit risk, reduce loan defaults, and improve portfolio performance.

🎯 Business Problem

Financial institutions face the challenge of balancing loan growth with effective risk management. Poor lending decisions can lead to:

- 📉 Increased loan defaults
- 💰 Financial losses
- ⚠️ Higher credit risk exposure
- 📊 Poor portfolio performance
- 💼 Reduced profitability

This project helps decision-makers identify credit risk patterns and improve lending strategies using data-driven insights.

🎯 Project Objectives

The project aims to:

- 📊 Assess customer credit risk
- 💳 Analyse loan approval and rejection trends
- 📈 Evaluate credit score distribution
- 💵 Examine income and loan amount relationships
- 👥 Analyse borrower demographics
- 📉 Identify factors contributing to loan defaults

- 📊 Develop an interactive executive dashboard

🗂️ Dataset

The dataset contains customer financial and demographic information, including:

- Age
- Gender
- Annual Income
- Credit Score
- Loan Amount
- Years of Experience
- City
- Education
- Employment Type
- Loan Approved

🛠️ Tools & Technologies

- 🗄️ SQL Server
- 📊 Microsoft Power BI
- 📈 DAX Measures
- 🧹 Data Cleaning
- 📉 Data Modelling
- 📋 Business Intelligence Reporting
- 📊 Dashboard Design

🔍 Project Workflow

1️⃣ Data Extraction (SQL)

- Queried customer loan records
- Selected relevant financial variables
- Validated dataset integrity

2️⃣ Data Preparation

- Removed inconsistencies
- Checked missing values
- Prepared data model
- Created calculated measures

3️⃣ Data Analysis

Analysed:

- Loan approvals
- Loan defaults
- Credit score distribution
- Income distribution
- Loan amount trends
- Employment analysis
- Debt-to-income ratio
- Interest rates

4️⃣ Dashboard Development

Built an interactive Power BI dashboard enabling stakeholders to explore loan performance and customer risk profiles using filters and KPIs.

📊 Dashboard KPIs

The dashboard provides key business metrics including:

- 👥 Total Customers
- 💰 Total Loan Amount
- 📉 Default Rate
- 📈 Approval Rate
- 💳 Average Credit Score
- 💵 Average Annual Income

📈 Dashboard Analysis

💳 Credit Score Analysis

Evaluates customer creditworthiness based on credit score ranges.

Business Value

- Improve lending decisions
- Reduce default exposure
- Enhance risk assessment

💰 Income Analysis

Examines borrower income levels.

Business Value

- Assess repayment capacity
- Support loan affordability analysis
- Improve lending policies

📉 Loan Default Analysis

Measures default trends across customer groups.

Business Value

- Identify high-risk borrowers
- Improve portfolio quality
- Reduce financial losses

🧑‍💼 Employment Analysis

Compares loan performance by employment status.

Business Value

- Evaluate employment stability
- Improve borrower assessment
- Reduce lending risk

💵 Loan Amount Analysis

Examines approved loan values.

Business Value

- Monitor lending exposure
- Improve portfolio management
- Support strategic planning

💡 Key Insights

The analysis revealed several important findings:

☑ Customers with higher credit scores consistently demonstrated lower default rates.

☑ Borrowers with higher annual income generally qualified for larger loan amounts and showed stronger repayment capacity.

☑ Debt-to-income ratio was an important indicator of potential default risk.

☑ Certain loan purposes carried higher default rates than others.

☑ Employment stability positively influenced loan approval outcomes.

☑ High-risk customer segments can be identified early using a combination of financial indicators.

📌 Business Recommendations

Based on the analysis:

✔ Prioritise applicants with strong credit scores during loan assessment.

✔ Strengthen affordability checks using debt-to-income ratios.

✔ Develop risk-based pricing strategies.

✔ Introduce enhanced monitoring for high-risk borrowers.

✔ Encourage early intervention for customers showing signs of financial distress.

✔️ Get information from customers about home ownership and analyse it. Homeowners generally presented lower credit risk than non-homeowners.

✔️ Regularly review lending policies using dashboard insights.

💼 Accounting & Business Impact

From an accounting and financial reporting perspective, effective credit risk assessment supports:

📉 Balance Sheet

- Lower impairment losses on loans and receivables
- Higher quality loan assets
- Improved asset valuation
- Reduced provision for bad debts

📊 Income Statement

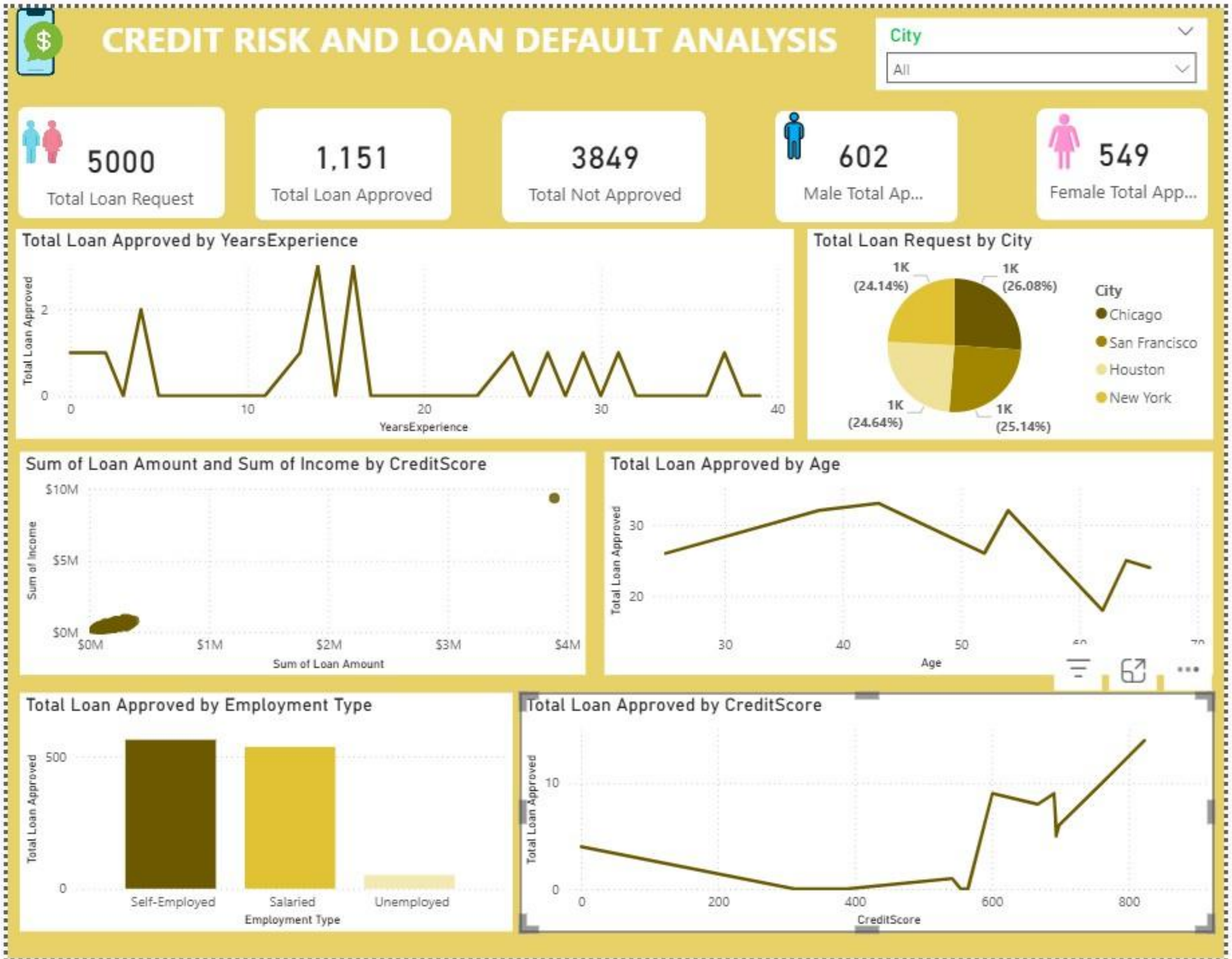
- Lower credit losses
- Improved net profit
- Increased interest income from performing loans

💰 Cash Flow

- Higher loan repayment rates
- Improved liquidity
- Better cash collection

Overall, robust credit risk management strengthens financial stability and supports sustainable business growth.

📷 Dashboard Preview




```
Images/
|── Credit Risk Dashboard.png
```

📁 Repository Structure

```
Credit-Risk-Assessment/
│
├── Dataset/
│   └── Credit Risk Dataset.csv
│
├── SQL/
│   └── Credit Risk Queries screenshot
│
├── Power BI/
│   └── Credit Risk Assessment.pbix
│
├── Images/
│   └── Dashboard Screenshot.png
│
├── Documentation/
│   └── Credit Risk Assessment Report.pdf
│
└── README.md
```

🚀 Skills Demonstrated

This project demonstrates proficiency in:

- 🗄 SQL Query Writing
- 📊 Data Cleaning
- 📈 Data Analysis
- 📉 Data Modelling
- 📊 Microsoft Power BI
- 📌 KPI Reporting
- 💳 Credit Risk Assessment
- 📉 Loan Default Analysis
- 💰 Financial Analytics
- 📊 Dashboard Design
- 📢 Business Storytelling
- 📋 Executive Reporting
- 🎯 Decision Support

🧑‍💻 About Me

Dayo Rex Afariogun

Data Analyst with practical experience in transforming raw data into actionable business insights using:

- 🗄 SQL
- 🐍 Python
- 📊 Microsoft Power BI
- 📈 Microsoft Advanced Excel

I have **B.Sc. in Accounting** and strong experience in **Business Intelligence, Financial Analytics, Commercial Analytics, Risk Analytics, and Data-Driven Decision Making**. I am actively seeking **Data Analyst opportunities in Ireland**, where I can apply my analytical skills to solve business problems and deliver measurable values.

📬 Connect with Me

- 💼 **LinkedIn:**
- 💻 **GitHub:** https://github.com/dayo-data-analyst
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⭐ Thank you for visiting this repository!

If you found this project interesting or helpful, please consider **starring ⭐ the repository** or connecting with me on LinkedIn or via Email. Feedback, suggestions are always welcome.